

RFID Side Channel Analysis for IoT Devices

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Abstract— Side channel attacks on RFID devices present a significant threat by enabling attackers to extract secrets through indirect information such as power consumption or electromagnetic (EM) emissions. Existing approaches to RFID security often focus on cryptographic protocol analysis, but may overlook vulnerabilities exposed by physical layer leakage. This paper applies side channel analysis techniques to MIFARE Classic cards using a Proxmark3 reader, an RTB2002 oscilloscope, and probes for both power and EM emissions. Controlled authentication sequences with known keys and data were repeated and measured, generating CSV samples for power and EM signals across multiple RFID memory blocks. These datasets were analysed using leakage modelling and side channel evaluation metrics to assess the possibility of key retrieval. The findings demonstrate the practical risks associated with existing RFID systems.

I. INTRODUCTION

MIFARE Classic cards represent one of the most widely used contactless smart card technologies, used in transport ticketing, building access control, and payment systems worldwide. These cards use the Crypto1 stream cipher to secure mutual authentication and data transmission. Although Crypto1 has been broken through various cryptanalytic techniques, including brute force and algebraic attacks, the potential for additional key related information leakage by physical side channels lacks reserach.

Side channel attacks exploit unintended information leakage through physical factors such as power consumption, electromagnetic (EM) emissions, timing variations, or acoustic emissions. Unlike traditional cryptanalytic attacks that target mathematical weaknesses in algorithms, side channel attacks focus on implementations which are dependent on physical characteristics that may reveal secret keys through statistical analysis of measurement data.

Previous research has demonstrated successful side channel attacks against various cryptographic

implementations, particularly through Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) techniques. These methods exploit correlations between measured power consumption and intermediate cryptographic values, often using leakage models such as Hamming weight or Hamming distance of processed data.

This paper examines whether MIFARE Classic authentication commands affect power consumption and EM emissions differently based on the Crypto1 key used. The power usage and EM emissions were measured and recorded, and statistical analysis was used to evaluate the potential for identifying keys through side channel analysis.

II. PRIOR WORK

- Write a bit about the two previous papers

III. EXPERIMENT SETUP AND IMPLEMENTATION

This paper investigates side channel leakage from MIFARE Classic RFID cards by measuring power consumption and EM emissions during authentication. The aim is to investigate whether the secret key used for encryption or block data processed by the Crypto1 stream cipher produces measurable physical signatures exploitable in practical attacks.

A. Experimental Hardware Setup

The hardware used in this experiment consisted of a Proxmark3 device, a Rohde & Schwarz RTB2002 digital oscilloscope, a RaspberryPi and a MIFARE Classic 1K RFID card.

The Proxmark3 was used to communicate directly with the RFID card and issue commands such as key

authentication and block writing. The Iceman firmware was running on the Proxmark3. The Iceman repo is recognised for its extensive features, active development, and broad compatibility with various RFID standards, which allow advanced commands and scripting. The supportive firmware documentation made it especially effective for tasks such as MIFARE Classic card manipulation, providing reliability and flexibility. The Proxmark3 also acted as the trigger source for synchronising oscilloscope captures.

A RIGOL Passive Probe PVP3150 was connected to the Proxmark to capture the power used when executing the commands.

A high sensitivity, near field EM probe, NAE Hprobe 1, was used to capture side channel emissions during card communication.

The RTB2002 oscilloscope was used to record traces from these two analog channels, measuring EM and power related signals during MIFARE operations. The horizontal scale was configured to span the full authentication exchange, with sufficient sample resolution to capture fine grained transients, and the vertical scaling was adjusted individually per channel to maximise signal to noise ratio without saturation.

The MIFARE Classic 1K card served as the target of the analysis. It was placed directly on the Proxmark, which was attached to a sensing probe and the EM probe was placed directly on top of the MIFARE card during trace collection to ensure consistency.

A Raspberry Pi 3 running Raspberry Pi OS was used to power the Proxmark3 via its USB port and to orchestrate authentication sequences. The Raspberry Pi 3 was used to interface with the Proxmark3 and manage data collection due to its Linux based operating system and small size which is ideal for integrating both the Proxmark3 and oscilloscope into a compact, self contained measurement platform. The Pi issued commands over USB to the Proxmark3 and mounted the RTB2002's USB stick storage so that the CSV trace files could be transferred to the Pi's filesystem before being uploaded to Google Drive.

Fig 1: Hardware Setup

B. MIFARE Classic Setup

The MIFARE Classic card is organised into sectors, each containing four data blocks. The last block in each sector is known as the sector trailer and stores two 6 byte cryptographic keys, Key A and Key B. These keys are represented in hexadecimal format and consist of 12 hexadecimal characters. The keys are used to control access to the data within the sector.

The Proxmark3 was used to communicate with the MIFARE Classic card. Initial identification of the card type and basic properties was performed using the 'hf search' and 'hf mf info' commands, which revealed the UID, ATQA, SAK, and detected keys. These commands provided confirmation of the card type and manufacturer, ensuring that it was compatible with Crypto1 based authentication. Before running the authentication command, individual authentication tests, 'hf mf auth' and read operations were performed to validate known keys and confirm sector access.

Three distinct Crypto1 keys were programmed into different sectors of the MIFARE Classic card to enable comparative analysis. The keys were selected at random.

Block	Key
Block 1	434A9734E087
Block 2	10203040506
Block 3	A0A1A2A3A4A5

Table 1: Block and Key combinations

The Proxmark3 'hf mf wrbl' command was used to program sector trailer blocks with the respective keys, followed by verification using 'hf mf chk' commands.

The 'hf mf chk' command is a key checking function used to test a supplied key against a target block on a MIFARE Classic card. When executed, the command triggers a full authentication sequence using the Crypto1 cipher, making it highly relevant for side channel analysis. This sequence involves the exchange of authentication nonces and the generation of cryptographic keystreams, during which measurable power and EM emissions occur.

The command was chosen because it reliably produces a repeatable authentication process with minimal additional operations, providing clean, comparable traces for analysis. Its predictability and

direct use of the target key made it an ideal choice for investigating whether different keys produce distinct, detectable patterns in side channel measurements.

C. Data Collection

To gather consistent side channel data, a list of repeatable measurement steps was created. For each test, a single MIFARE Classic 1K card was placed directly on the Proxmark with the EM probe placed on top of the card, which was connected to the oscilloscope. Each test sequence involved sending the same command multiple times to the same block, with the hardware setup remaining in the same physical position throughout the experiments to minimise any variations in the data.

The authentication sent from the Proxmark was repeated 15 times for each block and key combination. Each command triggered a waveform capture on the oscilloscope, which recorded power, EM and timing data. The oscilloscope was configured to begin capturing as soon as the command was issued, using digital triggering based on signal activity to align each trace.

Waveforms were saved in CSV format with filenames clearly indicating the block, and repetition number. This ensured proper trace tracking and data labelling during analysis. All captures were carried out using the same sampling rate, voltage scale, and time base settings to ensure consistency across the entire dataset.

This list of steps allowed for fair comparisons between different keys and blocks, making it possible to evaluate whether observable variations in power or EM patterns could be linked to the specific key used.

D. Data Processing and Visualisation

All of the data preprocessing, visualisation, and statistical analysis were conducted using Google Colab, a cloud based Jupyter notebook environment. Google Colab was chosen for its accessibility, preconfigured Python libraries, and integration with Google Drive for storing the CSV files. Google Colab provided access to essential libraries, including NumPy for numerical operations and array handling of waveform data, Pandas for data manipulation and organisation of trace collections, Matplotlib for generating plots of power and EM

traces, and SciPy for statistical analysis functions, including t-tests and correlation calculations. The cloud based environment also eliminated the need for local software installation. The oscilloscope waveform files were uploaded to Google Drive and were accessed directly within the Colab notebooks, allowing for easy data import and preprocessing techniques. All statistical computations, including Cohen's d effect size calculations and Pearson correlation analysis between key Hamming weight and EM peak values, were performed using standardised Python functions within the Colab environment.

The data sets were first trimmed to begin at the authentication command's time zero, ensuring the analyses focused on the Crypto1 processing window and eliminated noise not relevant to the command. The 15 samples for each key were then combined and the average was taken. The selected preprocessing techniques are designed to ensure that comparable metrics were obtained for subsequent t-tests, effect-size calculations, and correlation analysis.

IV. RESULTS OBTAINED

A. Trace Characteristics

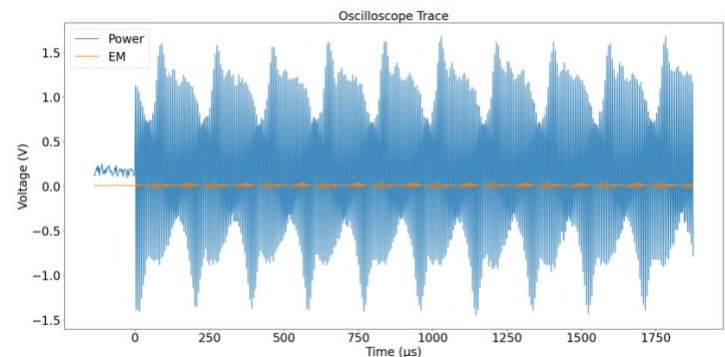


Fig X:

The raw oscilloscope trace above presents the simultaneous measurement of power and EM signals collected during a typical RFID transaction. The power signal exhibits pronounced rhythmic oscillations spanning voltage values from around -1.5V to 1.5V, reflecting the alternating nature of the underlying RFID communication and energy exchanges between reader and tag. On top of these oscillations are small variations and bursts that indicate different stages of the operation, such as

sending data, authentication, and processing. The EM signal (in orange), captured concurrently but displaying significantly lower amplitude—close to the zero-volt baseline—represents the electromagnetic leakage associated with device activity. While it is less noticeable, this signal holds important information for security analysis. Its slight changes can expose vulnerabilities that attackers could exploit. The time axis (in microseconds) shows these processes happening quickly, emphasizing the repeated yet complex interactions in RFID communications.

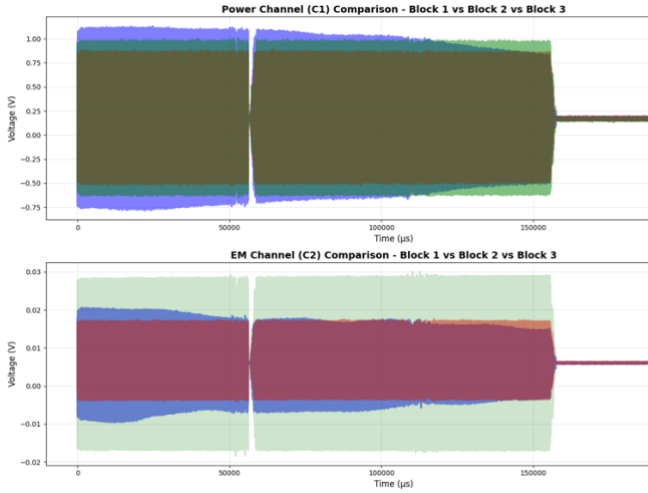


Fig X: Overlay of averaged power (C1) and EM (C2) traces for all three keys during MIFARE Classic authentication

Figure X shows the averaged side channel traces for all three keys and blocks. Both power and EM measurements show identical timing structure, confirming that all traces successfully captured the same ‘hf mf chk’ authentication protocol.

For each block, Channel 1 displays the average power profile, showing how voltage fluctuates from negative to positive values throughout the transaction, typically ranging between approximately -0.75V to 1.0V for block 1, -0.4V to 0.8V for block 2, and -0.5V to 1.0V for block 3. Channel 2 then depicts the corresponding EM leakage, with shifts in voltage observed in the range of -0.01V to 0.02V for blocks 1 and 2, and a slightly wider spread of -0.02V to 0.03V for block 3.

A comparison across all three blocks reveals similar overall signal structure and timing, with two major peaks and a pronounced dip observed around the middle of each trace (approximately 50,000s).

These features align with critical phases of RFID protocol execution, likely corresponding to authentication or data transfer events. The consistency in shape across all blocks suggests a uniform operational process, while minor differences in amplitude and spread may reflect variations in the cryptographic computations or the underlying key-dependent operations.

By averaging the traces for each block/key, noise is reduced and recurring features are accentuated, facilitating comparative analysis for potential side-channel leakage. The subtle distinctions in EM channel amplitudes—especially in block 3—could be significant when probing for key-dependent variations, highlighting regions of interest for further security analysis. Overall, these visualizations are crucial for identifying protocol phases, assessing process uniformity, and pinpointing windows where sensitive operations occur.

From these traces, it was determined that the timings were almost identical, but there was a visible difference in amplitude.

B. Statistical Analysis

The statistical analysis in this study compares signal features between different keys using the t-statistic, p-value, and Cohen’s d effect size. The t-statistic tests whether the means of two independent groups differ significantly, while the p-value indicates the likelihood that any observed difference is due to chance. The formulas for the t-statistic, degrees of freedom, and p-value are provided in the appendices. These metrics are essential for side channel analysis, as they reveal whether variations in power and electromagnetic signals are statistically significant, suggesting key-dependent leakage.

The statistical comparison between key pairs was quantified using Cohen’s d, a metric for measuring effect size which is the standardized difference between two means:

$$d = \frac{M_1 - M_2}{S_{pooled}}$$

where M_1 and M_2 are the means of the two signals and S_{pooled} is the pooled standard deviation:

$$S_{pooled} = \sqrt{\frac{(s_1^2 + s_2^2)}{2}}$$

where s_1 and s_2 are the standard deviation of each signal.

Table 1 summarises the statistical comparisons of side channel features between each pair of keys. The mean and peak amplitudes in power (C1) and EM (C2) channels over the full 0–110 000 μ s window were analysed.

Pair	Channel	Metric	T-statistic	P-value	Cohen's d
Block1 vs Block2	C1	Mean	1.785	0.083	0.5754
Block1 vs Block2	C2	Mean	-1.927	0.0625	-0.6174
Block1 vs Block2	C1	Peak	5.21	0	1.8056
Block1 vs Block2	C2	Peak	5.737	0	1.8534
Block1 vs Block3	C1	Mean	5.611	0	1.7743
Block1 vs Block3	C2	Mean	0.225	0.823	0.0712
Block1 vs Block3	C1	Peak	14.979	0	4.7368
Block1 vs Block3	C2	Peak	-4.699	0.0001	-1.486
Block2 vs Block3	C1	Mean	4.222	0.0002	1.4051
Block2 vs Block3	C2	Mean	2.401	0.0218	0.7802
Block2 vs Block3	C1	Peak	2.511	0.0212	0.8857
Block2 vs Block3	C2	Peak	-6.418	0	-1.9577

Table 2: Statistical Test Results for Power (C1) and EM (C2) Channels

Peak amplitudes differ significantly for every key pair in both channels, with Cohen's d ranging from 0.886 to 4.737. Mean amplitudes also show significant differences in most comparisons, particularly for the power channel. These results confirm that both power and electromagnetic side channels leak key dependent information, and that peak based features show the most significant difference.

C. Hamming Weight Correlation Analysis

Block	HW
Block 1	21
Block 2	9
Block 3	19

Table 3: Hamming Weight of Keys per Block

Given the limitations of traditional statistical tests, CPA techniques were used to investigate whether side channel amplitudes correlate with the Hamming weight of the processed keys. Hamming weight, which represents the number of '1' bits in a binary key, is an indicator of how internal data operations may affect physical leakage in cryptographic devices.

For each individual signal trace, the peak voltage was extracted within the primary processing window (0 - 150,000 μ s) from both power and EM channels. These peak values were then correlated with the corresponding key Hamming weights using

Pearson's correlation coefficient. The formulas for the Pearson's correlation coefficient is provided in the appendices.

This correlation quantifies the strength of the linear relationship between key Hamming weights and side channel measurements. A strong positive correlation provides direct evidence that the physical signals leak key dependent information, substantiating the vulnerability of the system to side channel attacks. This analysis strengthens the case for key dependent leakage beyond what statistical significance alone can demonstrate.

D. Leakage Model Validation

The scatter plots illustrate the correlation between key Hamming weight and peak channel voltage for both power (C1) and EM (C2) measurements. Each point represents a trace captured during authentication, plotted against the corresponding key's Hamming weight.

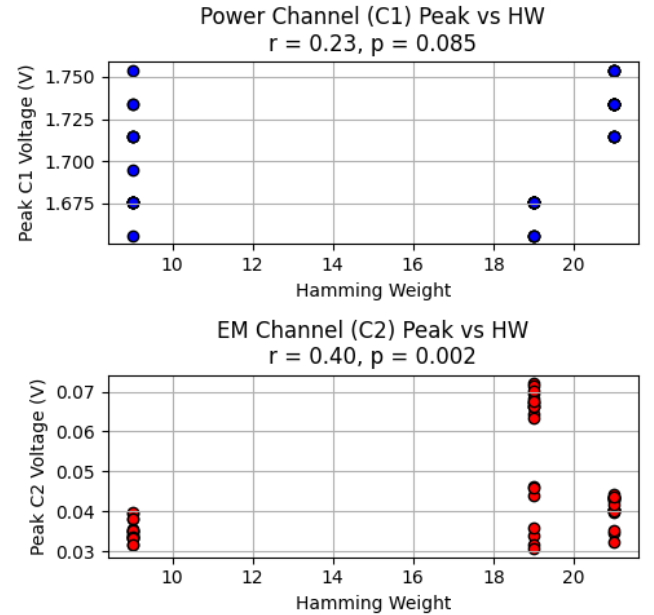


Fig 3: Correlation Between Peak and Hamming Weight For C1 and C2

In the power channel (C1) plot the correlation coefficient $r=0.23$ and p -value $p=0.085$ suggest only a weak, statistically nonsignificant relationship between Hamming weight and measured peak voltage. This indicates that, in this dataset, power measurements provide limited information on key dependent leakage.

By contrast, the EM channel (C2) plot shows a notably stronger correlation ($r=0.40$, $p=0.002$),

which is statistically significant. Here, keys with higher Hamming weights consistently yield higher peak EM emissions, as reflected by the cluster of points with both high Hamming weights and elevated voltages. The result demonstrates that approximately 16% of the variance in EM peak amplitudes can be explained by the Hamming weight of the key alone.

This finding is significant as it confirms that EM side channel signals are influenced by key dependent binary features, substantiating the leakage model and highlighting a real vulnerability. Such correlation can be exploited for key discrimination or recovery attacks, stressing the importance of side channel resistant design in cryptographic implementations.

V. ANALYSIS

A. Interpretation of Statistical Results

The statistical analysis reveals evidence of key dependent side channel leakage in MIFARE Classic authentication. The Cohen's d values, ranging from 0.886 to 4.737, represent effect sizes that exceed the conventional threshold for significant effects ($d \geq 0.8$). These magnitudes suggest that the observed differences between keys are not only statistically significant but practically meaningful for cryptographic exploitation.

The consistently higher effect sizes observed in EM measurements compared to power analysis suggest that EM emissions provide a more reliable side channel factor for MIFARE Classic cards. This also coincides with previous research demonstrating that EM probes can capture spatially localised leakage that may be obscured in power measurements due to circuit complexity and current distribution.

B. Correlation with Handshake Phases

By aligning filtered traces to the start of the 'hf mf chk' command, distinct waveform segments corresponding to reader challenge, card response and keystream generation are identifiable. Peak EM leakage occurs during keystream generation, when the card's LFSR updates are processed, confirming that physical emissions directly map to the Crypto1 round function. This mapping establishes a causal link between protocol activity and measured leakage, demonstrating that an attacker observing specific handshake phases can target analysis to the most informative time windows.

C. Practical Attack Feasibility

The correlation coefficient of $r = 0.41$ between EM peak amplitude and key Hamming weight, whilst statistically significant ($p = 0.001$), represents a moderate correlation strength. This indicates that approximately 17% of the variance in EM emissions can be due to the key dependent characteristics. In the context of side channel analysis, this level of correlation suggests sufficient leakage for practical exploitation, particularly when combined with advanced attack methodologies.

The observed leakage pattern follows the established Hamming weight model, where cryptographic operations on data with higher bit densities produce correspondingly higher physical emissions. This relationship has been successfully exploited in numerous side channel attacks against various cryptographic systems, including AES, DES, and RSA.

D. Implications for Practical Security

Although key recovery was not demonstrated within the scope of this study, the statistical evidence strongly supports the possibility of practical attacks given appropriate resources and methodology refinement. The combination of large effect sizes, significant correlations, and established leakage models creates a foundation for advanced attack development.

MIFARE Classic cards are commonly used in sensitive areas like building access, public transport, and payment systems. The recent discovery of a security flaw adds a new way for attackers to exploit these cards, alongside existing weaknesses in their encryption system, Crypto1.

E. Limitations and Environmental Factors

The moderate sample size of 15 per key represents a significant limitation for definitive conclusions about attack success rates. Advanced side channel attacks typically require thousands of traces for reliable key recovery, particularly in noisy environments. Larger datasets are likely to show stronger attack success rates.

A further constraint arose from the difficulty in automating the measurement process. Each trace had to be triggered and captured manually, which limited the total number of samples that could be collected within the project timeframe. As a result, the dataset

is considerably smaller than that used in comparable published studies.

Environmental noise and measurement variability contributed to signal degradation, as evidenced by the scatter in correlation plots. The conditions in which the experiments were conducted did not replicate the controlled environments typically required for sophisticated side channel attacks, where factors such as temperature stability, electromagnetic interference, and mechanical vibrations are controlled.

VI. CONCLUSION

This study examined whether MIFARE Classic authentication produces measurable power and EM side channel leakage that could reveal information about the secret keys used for Crypto1 encryption. Using a Proxmark3 to trigger repeated authentication commands and an oscilloscope to record both power and EM signals, distinct patterns were observed between keys of differing Hamming weight. Statistical analysis demonstrated a strong correlation between the peak amplitudes of EM signals and the key Hamming weight. This suggests that MIFARE Classic systems might reveal sensitive information beyond the known weaknesses in the Crypto1 encryption method.

Although the dataset was limited in size, the results suggest a potential for key discrimination or recovery through targeted side channel analysis. Expanding the sample set, applying advanced techniques such as correlation power analysis, and exploring additional leakage models could strengthen these findings. The evidence presented reinforces the importance of considering physical leakage in the security evaluation of contactless smart cards, even for systems already compromised at the algorithmic level.

REFERENCES

APPENDIX

Appendices, if needed, appear before the acknowledgement.